

ROBOC UPJUNIOR RESCUE LINE 2026

TEAM DESCRIPTION PAPER

Team Name: Bricks

Abstract

Our team developed an autonomous rescue robot capable of real-time navigation and victim rescue in complex environments. The main objective of this project was to create a robot that could accurately follow lines, detect rescue zones, and safely transport victims while overcoming obstacles commonly found in rescue environments. Since disaster-response scenarios often involve unstable terrain, obstacles, and limited accessibility, the robot had to be both mechanically durable and computationally reliable.

Unlike ordinary robots that focus only on movement, our robot was designed with both mobility and rescue capability in mind. The system combines stable hardware, sensor-based environmental recognition, and optimized software algorithms to perform rescue missions autonomously. In particular, the robot was developed to satisfy competition requirements while maintaining practical performance in challenging situations.

To improve mobility and reduce unnecessary weight, the robot uses medium motors instead of large motors, making the entire structure lighter without significantly sacrificing performance. By reducing total weight, the robot achieved better maneuverability and efficiency during long operations. At the same time, a gear reduction system was implemented to compensate for the lower motor power and ensure sufficient torque for lifting victims and climbing slopes.

The robot uses color sensors and a gyro sensor to recognize environmental conditions and maintain driving stability. Color sensors allow the robot to detect lines and identify rescue areas, while the gyro sensor continuously monitors the robot's tilt and angle. Through this system, the robot can adjust speed and maintain balance when climbing slopes or moving across uneven terrain. Additionally, a lifting mechanism powered by medium motors enables the robot to pick up and transport victims safely.

One of the most important strengths that distinguishes our robot from competing teams is the high reliability of line tracing achieved through repeated testing and optimization. In this competition, line tracing is the most heavily weighted scoring category because rescue cannot occur unless the robot successfully reaches the victim area. Recognizing this, our team prioritized stable movement and continuously improved line-following performance through repetitive testing. As a result of these improvements, the robot demonstrated highly stable movement and consistently reached victims without deviating from the intended path. During simulated competition tests, the robot achieved an average line deviation rate of less than 5%, proving its reliability and efficiency. These results demonstrate not only the robot's technical competitiveness but also our team's ability to solve engineering challenges through systematic testing and improvement.



1. Introduction

a. Team

Team Member: Taewook Lim (Hardware Development and Testing)

Taewook Lim is a first-year high school student with award-winning experience in the Korea RoboCup Competition. In this project, he was responsible for hardware development, structural design, and performance evaluation.

His role primarily focused on ensuring the robot's durability and operational stability. Since rescue robots often encounter walls, obstacles, and unstable environments, considerable attention was given to reinforcing the robot's structure. Through repeated testing, different hardware configurations were evaluated to determine the most effective balance between strength, weight, and mobility.

In addition to hardware construction, Taewook also conducted repeated performance evaluations to verify stability under various environmental conditions. Whenever problems such as wheel slippage, instability, or collision risks appeared, structural modifications were suggested and implemented. Furthermore, he contributed to project analysis by evaluating testing results and recommending improvements to maximize the robot's overall performance.

Team Member: Taehyung Lim (Software Development)

Taehyung Lim is also a first-year high school student with award-winning experience in the Korea RoboCup Competition. He was responsible for software development, sensor integration, and movement control programming.

One of his major responsibilities was strengthening the robot's line-tracing capability, which was considered the most critical aspect of the competition. Since rescue operations become impossible if the robot cannot reach the victim location, line-following stability was prioritized over all other functions.

To improve performance, he repeatedly tested and modified the software according to different situations and environmental conditions. Sensor threshold values were continuously adjusted to optimize movement accuracy. Whenever the robot deviated from the path or failed to recognize lines properly, corrective algorithms were introduced to recover navigation. As a result, the robot achieved significantly improved driving stability and competitive performance.

2. Project Planning

a. Overall Project Plan

The primary objective of this project was to develop a robot capable of stable line tracing and successful victim rescue while complying with competition regulations.

Several constraints had to be addressed during development.

Physical Constraints

The robot had to remain within competition size limits while maintaining enough durability and mobility to pass through slopes and obstacles safely. Since rescue areas included ramps and uneven terrain, stability during movement became a critical factor in hardware design.

Time Constraints

The development period was limited to one month, requiring efficient planning and strict milestone management. Each stage had to be completed within the scheduled timeline to ensure enough time remained for testing and improvement.

Competition Rules

Competition regulations limited sensor types, motor categories, and rescue methods. Therefore, the team carefully analyzed the rules and prioritized line-tracing performance, since reaching the rescue zone was essential for completing the mission.

Project Timeline and Milestones

December 20 – Mechanical Design and Construction Completed

Responsible: Taewook Lim (Hardware)

Tasks:

Motor placement optimization

Color sensor installation

Gyro sensor placement

Reinforcement of robot frame

December 27 – Line Tracing Software Development

Responsible: Taehyung Lim (Software)

Tasks:

Line-tracing algorithm implementation

Repeated testing and correction

Sensor threshold adjustment

January 10 – Final Optimization of Line-Tracing Program

Responsible: Entire Team

Tasks:

Final performance evaluation

Bug fixing and optimization before rescue programming

January 24 – Rescue Program Development

Responsible: Entire Team

Tasks:

Victim rescue area navigation

Rescue mechanism programming

Repeated execution and correction

The team followed a milestone order of mechanical design → software development → optimization → performance testing, because hardware completion was necessary before software testing could begin. Through repeated cycles of testing and revision, issues such as sensor malfunction and unstable navigation were gradually

3. Hardware

a. Mechanical Design and Manufacturing

The robot was designed with a reinforced outer frame to withstand collisions during rescue operations. In real rescue situations, robots may collide with walls or obstacles, so maintaining structural integrity is essential.

A four-wheel-drive system was implemented to maximize driving stability. Since all four wheels receive power simultaneously, the robot can maintain traction and balance even on rough surfaces or inclined terrain.

A lifting mechanism and platform were also installed to safely transport victims. The lifting system allows the robot to pick up and carry rescue targets while minimizing instability during movement.

Additionally, two medium motors were selected instead of larger motors to reduce overall weight. To compensate for lower power output, a gear reduction system increased torque, enabling the robot

b. Electronic Design and Manufacturing

The Port 1 color sensor, positioned at the very front of the robot, was designed to detect obstacles and identify rescue areas during navigation. Placing the sensor at the front enables the robot to recognize environmental conditions in real time before entering potentially dangerous areas. In rescue environments, unexpected obstacles such as walls, debris, or restricted pathways can interfere with movement, making early detection essential for stable operation. By continuously monitoring the surface and surrounding area, the Port 1 sensor helps the robot avoid collisions and maintain safe navigation. In addition, this sensor plays an important role in identifying designated rescue zones where victims may be located. Through color recognition, the robot can distinguish specific marked areas from ordinary paths, allowing it to transition smoothly from line tracing mode

into rescue mode. This significantly improves the robot's ability to search efficiently and respond accurately during rescue operations.

The Port 2 and Port 3 color sensors are responsible for the robot's line-tracing and navigation system. These sensors continuously detect and compare floor colors to determine whether the robot is correctly aligned with the designated path. Since line tracing is one of the most critical tasks in the competition, using two sensors instead of one greatly increases navigation accuracy and reduces the chance of losing the line. The robot determines its movement direction by comparing the values detected by both sensors. For example, when one sensor detects the black guiding line and the other detects the white floor, the robot recognizes that it has deviated from the center and automatically adjusts its direction. This enables the robot to perform left turns, right turns, stop when necessary, or continue moving straight while maintaining alignment with the path. Furthermore, repeated testing and calibration were conducted to optimize sensor thresholds under different lighting conditions and floor textures. As a result, the robot achieved highly stable navigation with an average line deviation rate of less than 5%, demonstrating the reliability of its line-following system. The Port 4 gyro sensor was installed to detect slope angles and changes in the robot's tilt, allowing the robot to maintain stability while traveling across uneven terrain. Since rescue environments often include ramps, inclined surfaces, or unstable pathways, maintaining balance becomes essential for successful operation. The gyro sensor continuously monitors the robot's orientation and provides real-time data about changes in angle and movement. When the robot begins climbing a ramp, the sensor detects the increased tilt and enables the control system to adjust motor speed accordingly. This prevents wheel slippage and ensures the robot climbs steadily without losing balance. Similarly, when descending a slope, the gyro sensor helps regulate speed to prevent sudden acceleration or instability. Through repeated slope-climbing tests, the robot successfully achieved a 100% success rate, demonstrating the effectiveness of the gyro sensor in improving movement

4. Software

c. General software architecture

The robot software architecture was designed around the process of Sensor Input → Data Analysis → Path Planning and Control → Actuator Movement, enabling the robot to autonomously recognize its environment, make decisions, and execute movement in real time. This structured approach allows the robot to efficiently process information from multiple sensors and respond appropriately to changing environmental conditions during rescue missions. Since rescue operations require both speed and accuracy, the software was carefully designed to ensure reliable communication between sensors, control algorithms, and movement systems.

The first stage, Sensor Input, is responsible for collecting environmental information through the robot's sensor system. The robot continuously receives real-time data from multiple sensors, including color sensors and the gyro sensor. The color sensors detect floor color differences, identify lines, recognize rescue zones, and detect obstacles in front of the robot. Meanwhile, the gyro sensor measures changes in tilt and orientation, allowing the robot to understand whether it is traveling on flat ground or climbing an inclined surface. By constantly gathering sensor data, the robot can maintain awareness of its surroundings and react immediately to environmental changes. Since sensor accuracy directly affects movement precision, repeated testing and calibration were performed to improve reliability under different lighting conditions and terrain types.

The second stage, Data Analysis, involves interpreting and processing the raw sensor information to determine the robot's current situation. Instead of simply reacting to individual sensor values, the robot compares multiple inputs simultaneously to make more intelligent decisions. For example, the software continuously analyzes readings from the Port 2 and Port 3 color sensors to determine whether the robot is properly aligned with the line. If one sensor detects the black line while the other detects the white floor, the robot recognizes that it has deviated from the center path. Likewise, the gyro sensor data is analyzed to determine whether speed adjustments are necessary while moving across slopes or uneven terrain. Through this analytical process, the robot transforms raw environmental data into meaningful information that supports decision-making.

The third stage, Path Planning and Control, is the core decision-making component of the robot software. Based on the analyzed sensor data, the robot determines what action should be performed next, such as moving forward, turning left, turning right, slowing down, or stopping. The control system uses predefined threshold values and conditional logic to ensure stable and predictable movement. In particular, line tracing was prioritized because successful navigation to the rescue area is essential for completing the mission. To improve navigation accuracy, the robot uses a line-tracing algorithm based on continuous sensor comparison and threshold evaluation. This allows the robot to make precise adjustments while following the designated path,

even in curved sections or intersections. Whenever the robot detects that it is beginning to deviate from the line, corrective actions are automatically applied to return it to the correct route.

One of the most important features of the software is its line recovery function, which was specifically designed to prevent navigation failure. During repeated testing, the team observed that even small sensor errors or environmental changes could occasionally cause the robot to drift away from the line. To solve this problem, an additional recovery algorithm was introduced. When deviation occurs, the robot automatically performs corrective movements by analyzing recent sensor data and adjusting motor direction until the line

is detected again. This feature significantly improved the robot's reliability and reduced the likelihood of failure during competition runs. As a result, the robot achieved an average line deviation rate of less than 5%, demonstrating strong navigation stability and accuracy.

The final stage, Actuator Movement, converts software decisions into physical actions through the robot's motors and mechanical systems. Once a movement command is generated, the motors execute the necessary actions, such as driving forward, turning, climbing slopes, or operating the lifting mechanism for victim rescue. Because the robot uses medium motors instead of larger motors, software optimization played an important role in maximizing efficiency while minimizing unnecessary energy consumption. Speed control was also carefully adjusted to ensure smooth movement and prevent instability, especially when climbing ramps or carrying victims. The lifting mechanism was integrated into the control system to allow the robot to safely pick up and transport victims after arriving at the designated rescue area.

A major advantage of this software architecture is its modular design, which allows each subsystem to be developed and tested independently while remaining easy to integrate into the complete robot system. For example, sensor processing, navigation algorithms, and motor control functions were implemented as separate modules, making it easier to identify errors and improve performance. This modular approach also increased maintainability, allowing future improvements or additional features to be implemented without redesigning the entire system. Through repeated testing, debugging, and optimization, the software architecture became increasingly reliable and efficient, contributing significantly to the overall success of the robot during performance evaluations.

4. Performance evaluation

Line-Tracing Stability Test

To evaluate the robot's navigation reliability, a line-tracing stability test was conducted in a simulated competition environment designed to closely resemble actual contest conditions. Since line tracing is one of the most important scoring criteria in the competition, ensuring stable and accurate movement was considered essential to the success of the project. The test environment included a variety of path conditions such as straight sections, curves, intersections, and sudden directional changes, all of which required the robot to continuously recognize and follow the designated line without interruption.

The robot completed a total of 20 repeated trials under identical conditions to verify consistency and minimize the possibility of random performance variation. During each trial, the robot's movement path was carefully observed to determine whether it could maintain proper alignment with the line while responding accurately to environmental changes. Special attention was given to curved paths and intersections, as these areas often cause navigation errors due to rapid directional adjustments.

To improve performance, repeated testing was followed by software adjustments and sensor recalibration. Sensor threshold values were optimized to ensure stable line detection under different lighting conditions and floor textures. In addition, a recovery algorithm was implemented to help the robot automatically return

to the intended path whenever slight deviations occurred. As a result of these improvements, the robot demonstrated highly stable navigation throughout the experiment.

Result: The robot achieved an average deviation rate of less than 5%, meaning that it rarely lost alignment with the line during operation. This result demonstrates the effectiveness of the robot's line-tracing system and confirms its ability to perform reliably in competition-like environments. The low deviation rate also highlights the importance of repeated testing and continuous software optimization in improving navigation performance.

Slope-Climbing Test

To verify the robot's ability to travel across uneven terrain, a slope-climbing test was performed using a 15-degree ramp, simulating the inclined surfaces commonly found in rescue environments. Since the competition environment includes sloped areas that robots must cross while maintaining balance and movement stability, this test was considered essential for evaluating mobility performance.

The robot completed 10 repeated climbing trials on the ramp to assess consistency and identify potential weaknesses in the driving system. During the test, the robot was required to ascend and descend the slope without slipping, losing balance, or experiencing movement failure. Particular focus was placed on whether the robot could maintain a stable speed and prevent wheel slippage while climbing.

5. Conclusion

Through this project, our team successfully developed an autonomous rescue robot capable of stable line tracing, obstacle navigation, and victim rescue in complex environments. The project was designed not only to satisfy competition requirements but also to simulate realistic rescue situations where robots must operate efficiently under difficult conditions. By integrating durable hardware structures, optimized sensor systems, and reliable software algorithms, we were able to create a robot that could perform rescue-related tasks autonomously and consistently.

One of the most important achievements of this project was the successful combination of hardware stability and software intelligence. The robot's reinforced frame and four-wheel-drive structure provided strong durability and balanced movement, allowing stable operation even when colliding with obstacles or traveling across uneven terrain. At the same time, the use of medium motors instead of larger motors reduced the overall weight of the robot while maintaining sufficient torque through a gear reduction system. This approach improved movement efficiency and enabled the robot to perform rescue actions such as lifting and transporting victims without sacrificing mobility.

The robot's sensor system also played a significant role in overall performance. Multiple color sensors enabled accurate line detection and rescue area recognition, while the gyro sensor improved driving stability by detecting slope angles and tilt changes in real time. Through continuous sensor monitoring and environmental analysis, the robot was able to adapt to different terrain conditions and maintain reliable navigation throughout testing. These features allowed the robot to operate autonomously without constant human intervention, demonstrating the effectiveness of the integrated sensor and control system.

A major focus of the project was improving the stability and reliability of the line-tracing system, since successful navigation is essential for reaching rescue areas. During early testing stages, the robot occasionally lost alignment with the path when traveling through curves or intersections. To solve this problem, our team repeatedly tested and modified both the software and hardware systems. Sensor threshold values were recalibrated, control algorithms were refined, and recovery functions were added to automatically return the robot to the intended path whenever deviation occurred. These improvements significantly increased navigation accuracy and reduced operational failure.

As a result of repeated testing and optimization, the robot achieved highly successful performance evaluation results. The line-tracing system recorded an average deviation rate below 5%, demonstrating accurate and

stable navigation. In slope-climbing experiments, the robot achieved a 100% success rate while traveling on a 15-degree ramp, proving the effectiveness of the gyro sensor and speed-control system. Furthermore, the rescue mechanism successfully completed repeated victim transportation tasks with a 90% rescue success rate, confirming the reliability of the lifting system and overall structural stability. These outcomes demonstrate that the robot was capable of performing both navigation and rescue functions efficiently in simulated competition environments.

Another important aspect of the project was the adoption of a modular software architecture. By separating the system into independent modules such as sensor processing, data analysis, navigation control, and actuator movement, the team was able to simplify testing, debugging, and maintenance. This modular structure also improved flexibility, allowing future upgrades and additional features to be implemented more easily. Through repeated experimentation and software refinement, the robot's overall performance became increasingly stable and adaptable to different environmental conditions.

Beyond technical achievements, this project provided valuable educational and collaborative experiences for all team members. Throughout the development process, the team strengthened important engineering skills such as problem-solving, critical thinking, creative design, and teamwork. Since unexpected technical issues frequently appeared during testing, team members had to analyze problems systematically and develop practical solutions under limited time and resource conditions. These experiences helped improve our ability to think logically and adapt quickly when facing engineering challenges.

The project also emphasized the importance of communication and cooperation within a team environment. Hardware and software development required continuous coordination between team members, and successful integration could only be achieved through active collaboration and shared decision-making. By dividing responsibilities efficiently and supporting one another during testing and troubleshooting, the team was able to maintain steady progress throughout the project timeline. This collaborative process not only improved the final robot design but also provided valuable experience similar to real-world engineering teamwork.

In addition, this project established a strong foundation for future robotics research and development. Although the robot successfully completed the required competition tasks, there are many opportunities for further improvement and expansion. For example, artificial intelligence and computer vision technologies could be integrated to improve victim recognition accuracy and environmental analysis. More advanced autonomous navigation algorithms could also allow the robot to operate effectively in more unpredictable and dynamic rescue situations. Furthermore, wireless communication systems and real-time monitoring functions could enhance practical usability in real disaster-response scenarios.

Ultimately, this project demonstrated that combining repeated testing, innovative design strategies, and strong teamwork can lead to the successful development of a reliable autonomous rescue robot. The knowledge and experience gained throughout this process will continue to support our future engineering and robotics projects. More importantly, the project showed how robotics technology can potentially contribute to solving real-world problems and improving safety in emergency rescue environments. Through this experience, our team gained not only technical skills but also confidence and motivation to pursue more advanced robotics development in the future.